Extremely Low-Cost Experimental Schemes for Neutrino Capture

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Abstract

Based on the geometric energy-flow model of the Unified Field Theory (UFT), this paper proposes three neutrino-capture experimental schemes with a total cost of less than RMB 300,000.

Unlike traditional large-scale water Cherenkov or liquid-scintillator detectors, these experiments are founded on the principle of geometric energy-flow perturbation, detecting the geometric effects of neutrino energy flow in ground laboratories through three independent methods:
(A) fiber-optic interference, (B) chiral torsion measurement, and (C) scintillation-array correlation.

Each scheme includes apparatus configuration, control requirements, data acquisition, and statistical criteria, and can be directly reproduced for experimental verification.

1. Theoretical Foundation

1.1 Geometric Model

According to Unified Field Theory, the universe is a right-handed helical energy-flow system expanding at the speed of light c.

Ordinary matter corresponds to right-handed closed-loop states, while neutrinos represent left-handed phase-inverted states of the same flow.

When a left-handed flow passes through right-handed spatial matter, the coupling is extremely weak due to geometric orthogonality, and interaction can occur only under local perturbation of the μ_g–Ω field direction.

Variations in neutrino flux induce:

$$\Delta\Phi_\nu \Rightarrow \Delta(\nabla \cdot (\mu_g \mathbf{v})$$

thereby generating minute optical, mechanical, or radiometric signals detectable by sensitive instruments.

2. General Experimental Design

Scheme	Detection Principle	Core Signal Type	Cost (RMB)	Reproducibility
A	Variation of circular	Optical	50 – 70 k	High
B	Chiral torque	Micro-torque	80 – 100 k	Medium
C	Scintillation-array	Count-rate	120 – 150 k	High

An experimental duration of at least 30 days is recommended to obtain statistically significant correlations with diurnal cycles and nuclear-reactor power variations.

3. Scheme A: Fiber-Optic Circular Birefringence Interference Experiment

3.1 Principle

When a left-handed neutrino flow passes through an optical fiber, the instantaneous perturbation of the spatial μ_g field induces a phase difference in the light beam:

$$\Delta\phi = \frac{2\pi L}{\lambda}(\Delta n_L - \Delta n_R)$$

where $\Delta n_L - \Delta n_R \approx 10^{-9} - 10^{-10}$ is within the detection limit of modern interferometers.

3.2 Apparatus Configuration

- Light source: Stable 1550 nm laser diode (power ≥ 5 mW, stability ≤ 10^{-4}).
- Optical fiber: Single-mode communication-grade fiber (total length ≥ 20 km, coiled on reels).
- Interferometer: Michelson configuration with high-stability reference arm.
- Detector: InGaAs photodiode + 16-bit ADC.
- Temperature control: Constant-temperature enclosure (drift ± 0.01 °C).

3.3 Operating Procedure

1. Fix the light source and interferometer on a vibration-isolated platform;
2. Adjust the arm-length difference to the midpoint of the interference fringe;
3. Begin data acquisition at 10 samples per second;
4. Record continuous data for ≥ 30 days.

3.4 Data Processing

- Extract the interference signal $S_3(t)$;
- Perform correlation analysis with the solar altitude angle h⊙;
- Primary criterion: $corr(S_3, h\odot) > 0.5$, $p < 10^{-6}$;
- Validation: Random nighttime phase controls should show zero correlation.

4. Scheme B: Chiral-Torque-Difference Experiment

4.1 Principle

The left-handed neutrino flow couples weakly to right-handed matter but can produce slight geometric resonance with left-handed structures.
When the neutrino flux varies daily or with nuclear-reactor power, a micro-torque difference Δτ appears between right-handed and left-handed samples.

4.2 Apparatus Configuration

- Torsion balance: optical-lever micro-torque meter (sensitivity ≤ 10^{-8} N·m)
- Suspension fiber: quartz, φ = 50 μm, length 50 cm
- Samples: left- and right-handed resin or quartz helices (equal mass, opposite chirality)
- Optical detection: laser + photodiode receiver
- Environment: sealed chamber, airflow < 0.1 cm/s, thermal drift < 0.05 °C

4.3 Procedure

1. Suspend left- and right-handed samples separately and record torsion angle $\theta(t)$;
2. Sample every 10 s, run continuously $\geq$ 30 days;
3. Swap sample positions weekly;
4. Use the same setup with achiral samples as a blank control.

4.4 Criteria

- Daily spectral-peak SNR > 5;
- Signal reverses sign after chirality swap;
- Statistical significance $p < 10^{-6}$.

Positive result: $\Delta\tau(t)$ shows a strong diurnal periodicity with opposite signs for opposite chiralities.
Negative result: $\Delta\tau$ is smooth noise without periodic features.

5. Scheme C: Scintillation-Array / Reactor-Power Correlation Experiment

5.1 Principle

Changes in reactor power cause neutrino-flux variation $\Delta\Phi_\nu$. If a left-handed energy flow passes through a scintillator array, it may trigger extremely low-probability flashes.

5.2 Apparatus Configuration

- Scintillator: BC-412 plastic blocks 10 × 10 × 2 cm × 9 units
- Photomultiplier: R7600U series
- Shielding: 3 cm lead + 5 cm polyethylene
- Data acquisition: multichannel counter (1 MHz sampling)
- Power stability: ± 0.1 %

5.3 Procedure

1. Place the array 20–30 m from a licensed reactor core;
2. Record count rate R(t) every 5 min;
3. Log reactor power P(t);
4. Compute the cross-correlation for the ON/OFF states.

$$C(\tau) = \int R\,(t)P(t+\tau)\,dt$$

5.4 Criteria

- Cross-correlation-peak lag < 30 min;
- Peak significance $p < 10^{-6}$;
- Results consistent after blind decoding.

6. Environmental Control and Blind-Test Procedure

Parameter	Requirement	Notes
Temperature stability	± 0.05 °C	Critical for A & B; use constant-temperature chamber or underground
Vibration noise	RMS < 10^{-7} m	Isolation table, higher weight for nighttime data
Electromagnetic shielding	Copper mesh + grounding	Suppresses 50 Hz interference
Blind protocol	Random file labeling	Analysts unaware of state labels
Data period	≥ 30 days	Ensures detection of diurnal cycle

7. Data Processing and Statistical Methods

1. Time-series analysis: FFT, power-spectral-density, daily-cycle peak detection.
2. Correlation analysis: Pearson and cross-correlation functions.
3. Significance evaluation: Monte Carlo shuffling (10^6 iterations) to compute p-values.
4. Multiplicity correction: Bonferroni adjustment to avoid false positives.
5. Reproducibility check: Verification across locations and sample geometries.

8. Expected Results and Scientific Significance

- If significant signals are observed, it will confirm that neutrinos are left-handed spatial energy flows and validate the μ_g–Ω geometric-perturbation mechanism.
- If no signal is detected, upper limits on $\Delta n, \Delta\tau, and\ \Delta\Phi_\nu$ will be established for calibrating future μ_g-field experiments.
- Overall cost is one-millionth that of traditional setups, suitable for university or civilian laboratories.

9. Safety and Ethics Statement

All experiments use no radioactive sources, relying solely on natural cosmic neutrino fluxes or licensed reactor-adjacent observations.
There are no biological hazards, and no high-voltage, high-temperature, or strong-radiation risks.

10. Conclusion

This study proposes extremely low-cost neutrino-capture experiments that realize detection based on μ_g–Ω field perturbation within the geometric energy-flow framework.
The three schemes are independent yet cross-verifiable, providing:

1. Replicable, low-threshold experimental routes;
2. Quantitative criteria ($p < 10^{-6}$) and a standardized statistical protocol;
3. A pathway for advancing Unified Field Theory from theoretical formulation to experimental physics.

Appendix — Budget and Execution Table

Scheme	Self-build Cost	Core Equipment	Duration	Main Criterion
A Fiber-optic interference	¥ 50–70 k	20 km fiber + interferometer	≥ 30 days	$\mathrm{corr}(S_3, h_\odot) > 0.5$
B Chiral-torque	¥ 80–100 k	Torsion balance + chiral samples	≥ 30 days	$\Delta\tau$ sign reversal, SNR > 5
C Scintillation array	¥ 120–150 k	Array + PMT	≥ 30 days	R–P cross-correlation $p < 10^{-6}$